



The triple oxygen isotope composition of the Earth mantle and understanding $\Delta^{17}\text{O}$ variations in terrestrial rocks and minerals

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ABSTRACT

It has been shown in numerous studies that terrestrial rocks and minerals fall on a line with slope ~ 0.52 in a $\delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ diagram. In this study, we present new data on the triple isotope composition of the Earth mantle, crustal materials and low- T aqueous precipitates. Crustal materials show distinct variations in $\Delta^{17}\text{O}$ suggesting that the concept of a single terrestrial mass fractionation is invalid on small-scale. Observable variations in individual fractionation slopes θ are interpreted as temperature effects. We show that the Earth mantle is isotopically homogeneous and confirm the recent finding that the Earth mantle has a negative $\Delta^{17}\text{O}$. The difference in $\delta^{18}\text{O}$ and $\Delta^{17}\text{O}$ between seawater and oceanic crust is discussed with respect to the effects of low- and high-temperature water–rock interaction.

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1. Introduction

1.1. Oxygen isotopes

Since the 1950s, it has been recognized that the fractionation between $^{17}\text{O}/^{16}\text{O}$ is close to one half that of $^{18}\text{O}/^{16}\text{O}$ (Craig, 1957), a feature recognized as mass dependent fractionation. It was realized that there was no need to measure the rare ^{17}O isotope of natural materials, because it provided no additional information than the $^{18}\text{O}/^{16}\text{O}$ ratio alone.

However, small, purely mass dependent deviations from a single fractionation line are seen, e.g. in water (Barkan and Luz, 2005; Landais et al., 2006, 2008; Luz and Barkan, 2010; Barkan and Luz, 2011). These variations are due to variable equilibrium and kinetic fractionation slopes in a $\ln(^{17}\text{O}/^{16}\text{O})$ vs. $\ln(^{18}\text{O}/^{16}\text{O})$ space (Eq. (3)). Such purely mass dependent variations in slopes of fractionation lines are predicted by theory (Young et al., 2002; Cao and Liu, 2011) and were anticipated by Matsuhisa et al. (1978) for rocks and minerals:

“Since the fractionation ratios, both for equilibrium and for kinetic processes, depend somewhat on the particular molecules or crystals involved, it must be expected that an unsystematic sampling of a large and complicates system, such as the Earth, will yield a slope on a graph of $\delta^{17}\text{O}$ vs $\delta^{18}\text{O}$ which is some average of the many individual slopes for specific processes.”

[Matsuhisa et al., 1978, p. 181]

Hence, it is not expected that rocks, minerals, and water samples fall on a single terrestrial fractionation line (TFL). In this study, we focus small mass dependent (opposed to mass independent variations, see Clayton, 2008, or Thiemens et al., 2012) deviations of rocks and minerals from a given linear relation in the $\ln(^{17}\text{O}/^{16}\text{O})$ vs. $\ln(^{18}\text{O}/^{16}\text{O})$ space.

1.2. Definitions

Variations in the oxygen isotope ratios are conventionally expressed in form of the δ -notation (McKinney et al., 1950) with:

$$\delta^{17}\text{O}_{\text{VSMOW}}^{\text{sample}} = 10^3 \left(\frac{\frac{^{17}\text{O}}{^{16}\text{O}}_{\text{sample}}}{\frac{^{17}\text{O}}{^{16}\text{O}}_{\text{VSMOW}}} - 1 \right) \quad (1)$$

and

$$\delta^{18}\text{O}_{\text{VSMOW}}^{\text{sample}} = 10^3 \left(\frac{\frac{^{18}\text{O}}{^{16}\text{O}}_{\text{sample}}}{\frac{^{18}\text{O}}{^{16}\text{O}}_{\text{VSMOW}}} - 1 \right) \quad (2)$$

Mass dependent fractionation (equilibrium or kinetic) between reservoirs (A, B) is defined by:

$$\alpha_{\text{A-B}}^{2/1} = (\alpha_{\text{A-B}}^{3/1})^{\theta_{\text{A-B}}} \quad (3)$$

The symbol $\alpha_{\text{A-B}}$ in Eq. (3) is the fractionation factor in $^{17}\text{O}/^{16}\text{O}$ (“2/1”) and $^{18}\text{O}/^{16}\text{O}$ (“3/1”) between reservoirs A and B. In case of equilibrium fractionation, A and B denote 2 phases (solids, liquids, and/or gases) that are in thermodynamic equilibrium. The variable $\theta_{\text{A-B}}$ is termed the triple isotope fractionation exponent. For oxygen, $\theta_{\text{A-B}}$ can vary between 0.5 and 0.5305 (Matsuhisa et al., 1978; Young et al., 2002; Cao and Liu, 2011).

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Table 1

List of sample names, descriptions and localities.

Sample	Description
SC	Mineral separates from the equilibrated San Carlos (33.39N, −110.53E, Arizona, USA) spinel lherzolite (olivine, orthopyroxene, clinopyroxene, spinel)
BIF	Mineral separates (quartz, hedenbergite, magnetite) from the equilibrated high-grade (~amphibolite facies) banded iron formation from Chennimalai (11.16N, 77.58E, southern India).
MMe	Orthopyroxene from mantle xenoliths from the Mount Melbourne Volcanic District (−74.40N, 164.69E, Victoria Land, western Antarctica)
MMa	Olivine from mantle xenoliths from the Meerfelder Maar (50.11N, 6.75E, Eifel, Germany)
Åh	Olivine from the dunite complex near Åheim (62.04N, 5.54E, Norway)
Ava	Olivine from a harzburgitic mantle xenolith from the Avachinsky volcano (53.15N 158.49E, Kamchatka)
AA	Garnet from the garnet lherzolite mantle xenolith from the Alpe Arami (46.23N, 8.98E, Swiss Alps)
GT	Plagioclase and quartz from a Paleoproterozoic granodiorite intrusion from the Västervik formation (57.85N, 16.62E, Sweden; Nolte, 2012)
UWG-2	Garnet reference material (Valley et al., 1995)
MORB	Mid ocean ridge basalt glass
Alt-OC	Altered oceanic crust sample
Dör	Quartz from the Dörentrup quartzite quarry (52.04N, 8.99E, Germany)
Chert	Siliceous nodular chert from the Cretaceous limestone cliffs near Étretat (49.71N, 0.21E, France)
Hex-Ros	Silica of hexactinellid sponge <i>Rosella cf. Antarctica</i> . Sample has been taken during the 2000 cruise ANT-XVII-3 of the RV Polarstern in the Southern Ocean (−72.29, −105.82, Antarctica) at a depth of 865 m. The growth temperature was $\sim 0.5 \pm 1^\circ\text{C}$ (Robertson et al., 2002) and the $\delta^{18}\text{O}$ of the seawater was $\sim 0\text{‰}$ (Meredith et al., 1999)
Demo	Silica of Demosponge sp. that was taken during the 2006 cruise M70/1 of the RV Meteor in the Adriatic Sea (41.51N, 17.69E, near Bari) a depth of 650 m. The growth temperature was $\sim 12.5 \pm 1^\circ\text{C}$ (Oddo and Guarnieri, 2011) and the $\delta^{18}\text{O}$ of the seawater was 1.5‰ (Stenni et al., 1995)
Homo-Plak	Sponge (Homoscleromorpha, <i>Plakortis simplex</i>) silica sample taken during the 2005 cruise of the RV Poseidon in a depth of ~ 300 m west of Norway (64.05N, 7.36E, Sula Reef Complex). The estimated growth temperature is $7 \pm 2^\circ\text{C}$
Hex-Eup	Sample Eup-sp is a hexactinellid sponge <i>Euplectella aspergillum</i> (Philippines; Owen, 1841). It was commercially acquired. The estimated growth temperature is $9 \pm 5^\circ\text{C}$ and the $\delta^{18}\text{O}$ of the seawater of $\sim 0\text{‰}$

For better illustration, small deviations in $^{17}\text{O}/^{16}\text{O}$ from a given reference line (RL) are expressed in form of the $\Delta^{17}\text{O}$ notation (Eq. (4)). We stress out here that $\Delta^{17}\text{O}$ itself is not a measured quantity. The measured quantities are $\delta^{17}\text{O}_{\text{VSMOW}}$ and $\delta^{18}\text{O}_{\text{VSMOW}}$ (Eqs. (1) and (2)).

$$\Delta^{17}\text{O} = 10^3 \ln \left(\frac{\delta^{17}\text{O}_{\text{VSMOW}}}{10^3} + 1 \right) - 0.5305 \times 10^3 \ln \left(\frac{\delta^{18}\text{O}_{\text{VSMOW}}}{10^3} + 1 \right) \quad (4)$$

Because $\Delta^{17}\text{O}$ is not a measured quantity it is somewhat arbitrary which slope of the RL (λ_{RL}) is chosen for defining $\Delta^{17}\text{O}$. Comparing data between laboratories requires recalculation of $\Delta^{17}\text{O}$ from the measured quantities (i.e. $\delta^{17}\text{O}_{\text{VSMOW}}$ and $\delta^{18}\text{O}_{\text{VSMOW}}$). In this contribution, we choose $\lambda_{\text{RL}} = 0.5305$ and $\gamma_{\text{RL}} = 0$ (intercept of the RL; see also [Wiechert et al., 2004](#)). The $\delta'^x\text{O}$ ($x = 17, 18$) notation in Eq. (4) is the linearized form of the δ -notation ([Miller, 2002](#)). The chosen RL slope should not be confused with $\theta_{\text{A-B}}$ (Eq. (3)) as λ_{RL} has no physical meaning.

The T dependence of equilibrium fractionation can lead to $\theta_{\text{A-B}} < 0.5305$ (e.g., [Matsuhisa et al., 1978](#); [Young et al., 2002](#); [Cao and Liu, 2011](#)). Variations in θ and $\delta^{18}\text{O}$ consequently lead to purely (or “canonical”, [Eiler, 2013](#)) mass dependent variations in $\Delta^{17}\text{O}$.

1.3. Terrestrial waters, rocks and minerals

Meteoritic waters fall on a line (MWL) with slope $\lambda_{\text{MWL}} = 0.528$ and $\gamma_{\text{MWL}} = 0.033\text{‰}$ ([Barkan and Luz, 2005](#); [Luz and Barkan, 2010](#)). The MWL is result of combined equilibrium (evaporation, condensation; $\theta_{\text{water-vapor}} = 0.529$, [Luz and Barkan, 2010](#)) and kinetic (diffusion; $\theta_{\text{diffusion}} \approx 0.518$) fractionation. Note that seawater (and VSMOW) does not fall on the MWL ($\gamma_{\text{MWL}} = 0.033\text{‰}$; [Barkan and Luz, 2011](#)). Hence, interaction between rocks and meteoric water is expected to impose a $\Delta^{17}\text{O} \neq 0$.

For rocks and minerals, a number of reference lines with slopes $0.5 \leq \lambda_{\text{TFL}} \leq 0.526$ have been proposed (e.g., [Clayton and Mayeda, 1978](#); [Miller et al., 1999](#); [Greenwood et al., 2005](#); [Spicuzza et al.,](#)

[2007](#); [Rumble et al., 2007, 2013](#); [Pack et al., 2007](#); [Kusakabe and Matsuhisa, 2008](#); [Ahn et al., 2012](#); [Tanaka and Nakamura, 2013](#)). With exception of [Tanaka and Nakamura \(2013\)](#), all studies assumed that the TFL passes through the origin in a $\delta^{17}\text{O}_{\text{VSMOW}}$ vs. $\delta^{18}\text{O}_{\text{VSMOW}}$ diagram. By carefully calibrating their reference O_2 relative to O_2 liberated from VSMOW, [Tanaka and Nakamura \(2013\)](#) resolved an intercept of $\gamma_{\text{TFL}} \approx -0.1\text{‰}$. Because the $\delta^{17}\text{O}_{\text{VSMOW}}$ of mineral reference materials (e.g., NBS-28 quartz, San Carlos olivine, or UWG2 garnet) is not known, often O_2 reference gas was “calibrated” relative to calculated $\delta^{17}\text{O}$ values of minerals (e.g., [Miller et al., 1999](#); [Spicuzza et al., 2007](#)). The calculations of $\delta^{17}\text{O}$ were based on the erroneous assumption that VSMOW $\in$ TFL. Complex recent and future applications of triple isotope measurements of terrestrial samples (solids, water, gases), e.g., as paleo- CO_2 proxy ([Bao et al., 2008](#); [Pack et al., 2013](#)), however, clearly need $\delta^{17}\text{O}$ reported on the same scales. For both, $\delta^{17}\text{O}$ and $\delta^{18}\text{O}$, only the VSMOW scale should be used.

Aim of this study is to present a framework on utilizing variations in $\Delta^{17}\text{O}$ of terrestrial rocks and to critically revisit the concept of a TFL. We present data on the composition of the Earth's upper mantle, demonstrate variations in equilibrium θ for different systems, and discuss implications for the triple oxygen isotope relation between the mantle, ocean crust, and seawater.

2. Materials and methods

2.1. Rocks and minerals

Sample names and sample descriptions are summarized in [Table 1](#).

2.2. Oxygen isotope analysis

The oxygen isotope analyses have been conducted by means of laser fluorination in combination with multicollector gas source mass spectrometry ([Sharp, 1990](#)). Sample oxygen was liberated by reacting about 2–3.5 mg silicates and oxides with ~ 60 mbar purified F_2 ([Asprey, 1976](#)). The samples were heated with a 50 W CO_2 laser. The liberated oxygen was purified from excess F_2 by reaction with NaCl ($T \approx 150^\circ\text{C}$) and removal of Cl_2 in a liquid nitrogen (LN) cold trap. Sample O_2 was then collected in a U-trap filled with 5 Å molecular sieve held at LN temperatures. After all sample

Table 2
Results of oxygen isotope analyses.

Sample	Type	$\delta^{17}\text{O}_{\text{VSMOW}}$ (‰)	SD	SE	$\delta^{18}\text{O}_{\text{VSMOW}}$ (‰)	SD	SE	$\Delta^{17}\text{O}$ (ppm)	SD	SE	N
<i>Mantle xenoliths</i>											
SC	ol	2.69	0.08	0.01	5.28	0.16	0.03	−103	8	1	35
SC	opx	3.09	0.11	0.06	6.03	0.22	0.12	−103	20	10	3
SC	cpx	2.93	0.07	0.05	5.72	0.13	0.01	−95	7	5	2
SC	spl	2.23	0.11	0.06	4.39	0.22	0.11	−96	12	6	4
AA	gt	2.86	0.10	–	5.59	0.20	–	−97	6	–	1
MMA	ol	2.63	0.05	0.02	5.16	0.10	0.05	−100	14	7	4
Ava	ol	3.01	0.10	–	5.86	0.20	–	−98	4	–	1
MMe	ol	2.81	0.11	0.05	5.49	0.06	0.03	−98	6	3	4
Åh	ol	2.88	0.10	–	5.61	0.20	–	−98	9	–	1
Weighted average								−101	3		55
<i>Oceanic crust</i>											
MORB	glass	2.86	0.10	0.04	5.60	0.18	0.07	−100	14	6	6
Alt-OC	bulk	6.62	0.10	–	12.78	0.20	–	−140	10	–	1
<i>Continental crust</i>											
NBS-28	qz	5.06	0.11	0.03	9.75	0.20	0.06	−104	8	2	13
Dör	qz	6.19	0.08	0.03	11.96	0.17	0.06	−134	6	2	9
Chert	qz	17.54	0.10	0.06	33.75	0.21	0.10	−227	8	5	2
BIF	qz	9.58	0.08	0.04	18.47	0.15	0.08	−176	10	6	3
BIF	hed	8.26	0.10	–	15.97	0.20	–	−175	4	–	1
BIF	mt	5.40	0.10	–	10.49	0.10	–	−156	6	–	1
UWG-2	gt	3.06	0.07	0.03	5.99	0.13	0.06	−102	7	3	5
GT	plag	4.48	0.05	0.03	8.68	0.10	0.06	−110	5	3	2
GT	qz	5.43	0.05	0.03	10.47	0.10	0.06	−115	5	3	2
<i>Sponge spicules</i>											
Hex-Ros	SiO ₂	18.3	0.5	–	35.6	0.9	–	−400	70	–	1
Demo	SiO ₂	19.4	0.5	0.3	37.6	0.9	0.5	−310	70	40	2
Homo-Plak	SiO ₂	17.9	0.5	0.3	34.7	0.9	0.5	−370	70	40	2
Hex-Eup	SiO ₂	17.7	0.5	0.3	34.1	0.9	0.5	−230	70	40	2

O₂ was collected, we released the sample by heating and transported the sample with a He carrier gas into a second in-stream 5 Å molecular sieve trap. This trap aids to cryofocus the sample for the next purification step. The O₂ was then back-released into the He carrier gas by heating in a water bath held at 90 °C and then passed through a ~10 m long 5 Å molecular sieve lined capillary gas chromatography column operated at room temperature. The purified O₂ was collected on a third in-stream 5 Å molecular sieve trap, from which sample O₂ was emitted to the variable bellow of a Thermo MAT 253 gas source mass spectrometer.

Samples were analyzed for ~90 min relative to bottle reference O₂. To ensure that the $\delta^{17}\text{O}$ of the reference O₂ is known relative to VSMOW, its isotope composition was measured by E. Barkan (The Hebrew University, Jerusalem) relative to VSMOW. We have confirmed the calibration by own analysis of VSMOW liberated by fluorination in Ni reactors using BrF₅ as reaction agent ($T = 300^\circ\text{C}$, 3 h). Our preliminary data agree within <10 ppm in ($\Delta^{17}\text{O}$) with the calibration by E. Barkan. We have also analyzed air O₂. The air O₂ was separated from Ar by means of low T gas chromatography (Yeung et al., 2012). Our datum falls between the numbers given by Barkan and Luz (2011) and Yeung et al. (2012). Both, VSMOW and air O₂ data show that our reference gas was calibrated correctly on VSMOW scale. Details of the VSMOW and air analyses will be presented elsewhere.

We report our $\delta^{17}\text{O}$ and $\delta^{18}\text{O}$ values in ‰ (parts per thousand, relative to VSMOW) and $\Delta^{17}\text{O}$ values either in ‰ (parts per thousand) or ppm (parts per million, molar isotopic ratios). We report 1 σ uncertainties in the text, illustrations, and tables. Given that the $\Delta^{17}\text{O}$ values for our reference materials agree with the values of Tanaka and Nakamura (2013) we suggest that the accuracy of our calibration is 10 ppm or better. Our MAT 253 is currently exclusively used for O₂ analysis. The external reproducibility of a single measurement was <0.1‰ for $\delta^{17}\text{O}$ and $\delta^{18}\text{O}$ and ~5 to 8 ppm for $\Delta^{17}\text{O}$ and is only slightly higher than the mass spectrometer noise. This is comparable to what is obtained by Barkan and Luz

(2005), Landais et al. (2008), or Luz and Barkan (2010) for water analyses.

Analyses of the sponge spicules were obtained by a previously developed technique, where we measured the oxygen in continuous mode (Gehler et al., 2011; Hofmann et al., 2012; Pack et al., 2013) with a precision in $\delta^{18}\text{O}$ of 0.1 to 0.2‰ and in $\Delta^{17}\text{O}$ of ± 40 ppm. The sponge spicules were analyzed along with NBS-28 quartz that was used as reference for $\Delta^{17}\text{O}$ (NBS-28 data from this study and Tanaka and Nakamura, 2013).

The regression lines and error envelopes were computed by Monte Carlo simulation using the MATHEMATICA (Wolfram Research Inc.) software.

3. Results

The oxygen isotope data are listed in Table 2.

3.1. Igneous and high- T metamorphic rocks

Olivine from the San Carlos lherzolite has a mean $\Delta^{17}\text{O}$ of -0.102 ± 1 ppm (Table 2). The San Carlos minerals (olivine: ol, orthopyroxene: opx, clinopyroxene: cpx, spinel: spl) fall on a mass fractionation line (MFL) with slope $\theta_{\text{spl-ol-opx-cpx}} = 0.529 \pm 0.006$ (Fig. 1A). Mantle opx (Avachinsky, Kamchatka), ol (Mt. Melbourne, Victoria Land, Antarctica; Meerfelder Maar, Eifel; Åheim, Norway), and gt (Alpe Arami, Alps) fall in a narrow range of $-100 \leq \Delta^{17}\text{O} \leq -97$ ppm (Table 2; Fig. 1A, open brown squares). Our sample of MORB glass has a $\delta^{18}\text{O}$ of 5.6‰ and a $\Delta^{17}\text{O}$ of -100 ± 6 ppm (Fig. 1A). A single sample of an altered oceanic basalt has a $\delta^{18}\text{O} = 12.7\text{‰}$ and a $\Delta^{17}\text{O} = -140 \pm 10$ ppm.

The UWG-2 garnet has a $\delta^{17}\text{O} = 3.06 \pm 0.03\text{‰}$ and $\delta^{18}\text{O} = 5.99 \pm 0.06\text{‰}$ (SE, $N = 5$). The corresponding $\Delta^{17}\text{O}$ is -102 ± 3 ppm. For NBS-28 quartz (qz), values of $\delta^{17}\text{O} = 5.06 \pm 0.03\text{‰}$, $\delta^{18}\text{O} = 9.75 \pm 0.06\text{‰}$ and $\Delta^{17}\text{O} = -104 \pm 2$ ppm were measured (SE, $N = 13$, Table 2). The $\Delta^{18}\text{O}_{\text{NBS-28-UWG-2}}$ ($\delta^{18}\text{O}_{\text{NBS-28-}}$

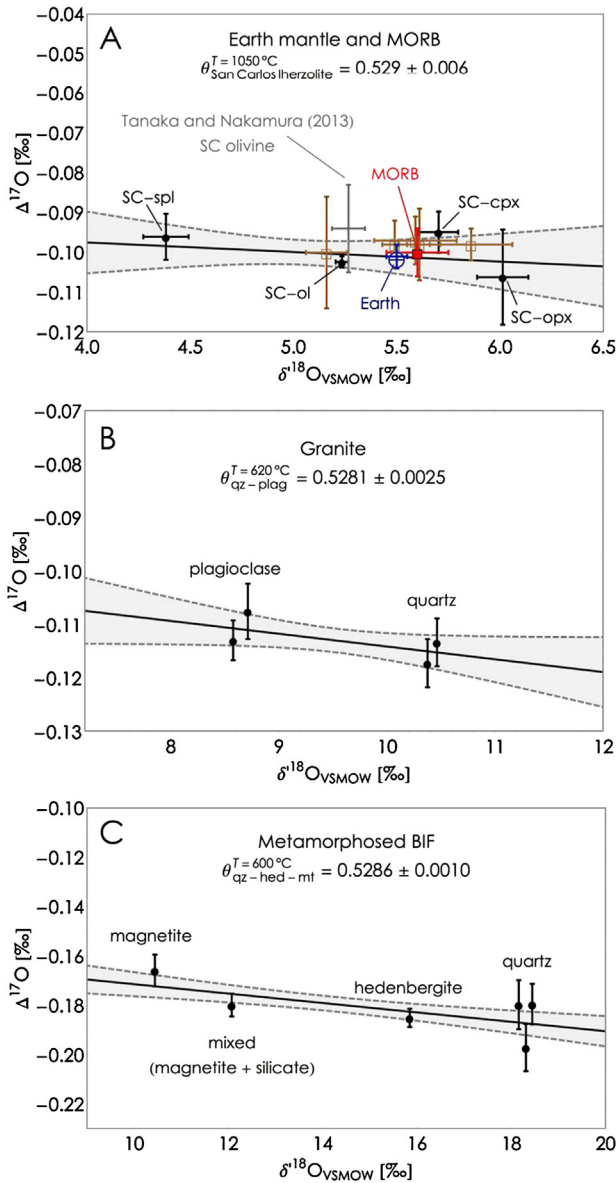


Fig. 1. Plot of $\Delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ of minerals from high- T igneous and metamorphic rocks (A–C) and MORB glass (A). The slopes of the regression lines are indicated. For comparison, the San Carlos olivine value of Tanaka and Nakamura (2013) is displayed (A). (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

$\delta^{18}\text{O}_{\text{UWG-2}}$ is 3.76‰. The inhouse qz reference material from the Dörentup quartzite has $\delta^{18}\text{O} = 11.89\text{‰}$ and $\Delta^{17}\text{O} = -134 \pm 2$ ppm.

Quartz and plagioclase (plag) of the granite sample have $\Delta^{17}\text{O} \approx -110$ ppm and define an MFL line with a slope of $\theta_{\text{qz-plag}} = 0.5281 \pm 0.0026$ (Fig. 1B). The plagioclase contains ~77 mol% anorthite.

The minerals (qz, hedenbergite: hed, magnetite: mt) in the metamorphosed BIF fall on a line with slope $\theta_{\text{qz-hed-mt}} = 0.5286 \pm 0.0010$. The average $\Delta^{17}\text{O}$ is at -180 ppm (Fig. 1C).

3.2. Low- T silica samples

The chert sample has a $\delta^{18}\text{O} = 33.2\text{‰}$ and a $\Delta^{17}\text{O}$ of -227 ± 5 ppm (Table 2). With water composed of a mixture of fresh- and seawater (see Discussion) it defines a line that has a slope of $\theta_{\text{chert-water}} = 0.5235 \pm 0.0002$ (Fig. 2A).

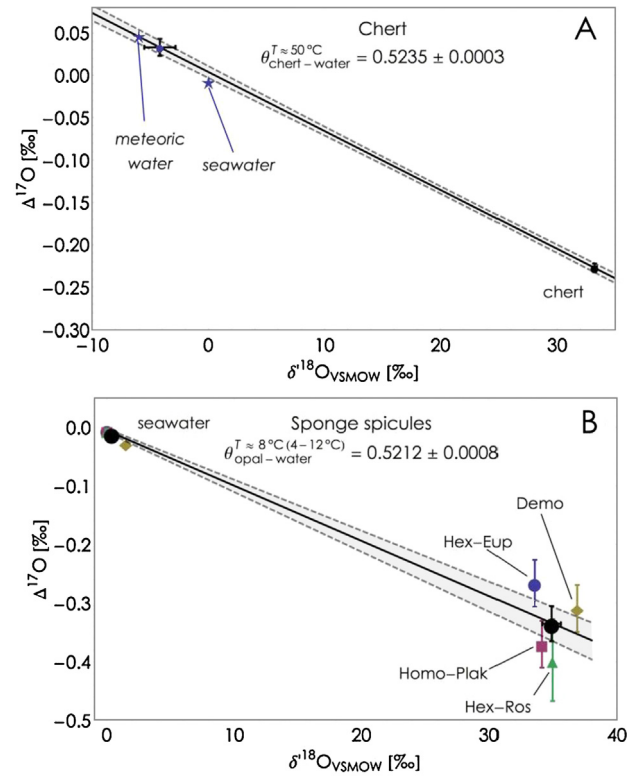


Fig. 2. Plot of $\Delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ of chert (A) and sponge (B) silica. The slopes of the regression lines are indicated.

Modern sponge silica is by ~35‰ enriched in $\delta^{18}\text{O}$ relative to the water it formed in. The $\Delta^{17}\text{O}$ varies between -390 and -270 ppm (Table 2). Pooling sponge spicule and water data gives a mean $\Delta^{17}\text{O} = -340 \pm 30$ ppm and a $\theta_{\text{silica-water}} = 0.5211 \pm 0.007$ (Fig. 2B).

4. Discussion

Before discussing our result with respect to geochemical processes, we compare our results with published data and discuss the concept of the TFL.

Our $\Delta^{17}\text{O}$ data are systematically lower than previously reported (Fig. 3). A reason for the discrepancy is that some studies did not report their $\delta^{17}\text{O}$ on VSMOW scale. An example is the study by Spicuzza et al. (2007). They calculated the $\delta^{17}\text{O}$ of UWG-2 to 3.046‰, which gives an apparent $\Delta^{17}\text{O}$ of -27 ppm (Eq. (4)). This is in contrast to our measured $\Delta^{17}\text{O}_{\text{UWG-2}} = -102 \pm 3$ ppm. The difference is due to the fact, that Spicuzza et al. (2007) assumed that their TFL passes through the origin (see also Miller et al., 1999). Those studies that explicitly report their $\delta^{17}\text{O}$ on VSMOW scale (Robert et al., 1992; Kusakabe and Matsuhisa, 2008; Ahn et al., 2012; Tanaka and Nakamura, 2013), however, also show a systematic shift toward higher $\Delta^{17}\text{O}$ (Fig. 3). This could be due to contamination of the sample O_2 gas by remaining traces of NF_3 (e.g., Clayton et al., 1976; Clayton and Mayeda, 1983; Rumble et al., 1997; Pack et al., 2007). We argue that gas chromatographic separation of NF_3 (see Materials and methods) is more efficient than cryogenic methods.

Our mantle xenolith data suggest that the Earth mantle is isotopically homogeneous with a mean $\Delta^{17}\text{O}$ value of -101 ± 2 ppm ($\delta^{18}\text{O}_{\text{VSMOW}} = 5.5\text{‰}$, Matthey et al., 1994, $\delta^{17}\text{O}_{\text{VSMOW}} = 3.01\text{‰}$). Using a more common slope of $\lambda_{\text{RL}} = 0.5247$ for the RL (with $\gamma_{\text{RL}} = 0$) gives $\Delta^{17}\text{O} = -69$ ppm for the mantle. Published high-precision data on $\Delta^{17}\text{O}$ of meteorites that were not reported for $\delta^{17}\text{O}$ on VSMOW scale (e.g., Franchi et al., 1999;

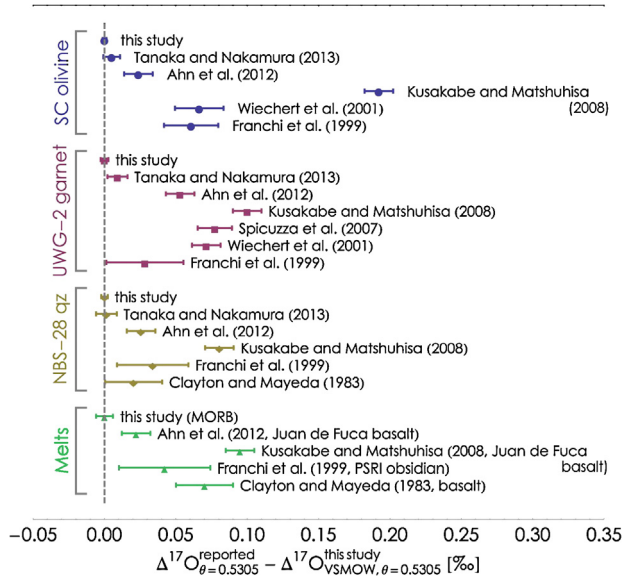


Fig. 3. Comparison of data from this study with literature data ($\Delta^{17}\text{O}$ recalculated using Eq. (4)). Our data agree well with the numbers published by Tanaka and Nakamura (2013), but are systematically lower than data from other studies (Clayton and Mayeda, 1983; Franchi et al., 1999; Wiechert et al., 2001; Spicuzza et al., 2007; Kusakabe and Matshuhisa, 2008; Ahn et al., 2012).

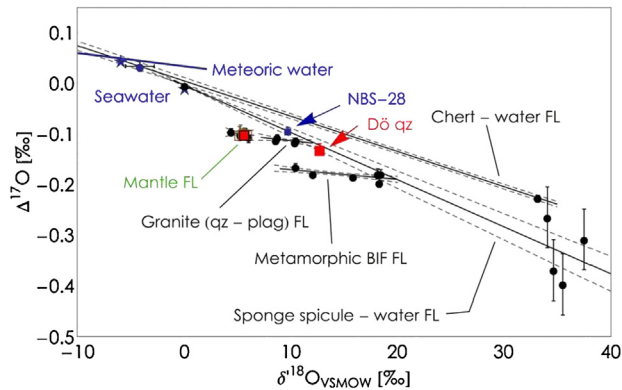


Fig. 4. Plot of $\Delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ showing the different MFLs in a small selection of terrestrial reservoirs. The data show that there is not a single TFL.

Wiechert et al., 2001, 2004; Greenwood et al., 2005; Spicuzza et al., 2007; Scott et al., 2009) should be corrected accordingly: $\Delta^{17}\text{O}_{\text{HED}} = -329 \pm 2$ ppm, $\Delta^{17}\text{O}_{\text{angrites}} = -164 \pm 3$ ppm, and $\Delta^{17}\text{O}_{\text{Mars}} = 250 \pm 14$ ppm.

Our recommended values for NBS-28 qz, UWG-2 gt, and San Carlos ol are listed in Table 2. These can be used to ensure the best interlaboratory comparability.

Our new data on terrestrial rocks and minerals clearly show that resolvable variations exist in $\Delta^{17}\text{O}$. These variations are irrespective of the chosen RL. Cogenetic phases (minerals, fluids) define their own MFLs (see Fig. 4 for a summary). This confirms the prediction of Matsuhisa et al. (1978) that there is not a single TFL. We thus suggest to abandon the concept of a rocks and minerals defined TFL.

In our Fig. 1A, C we suggest that >2 phases (e.g., spl, ol, opx, and cpx in the San Carlos Iherzolite) fall on a common “rock” MFL. In case of the San Carlos Iherzolite (Fig. 1A), however, actually 6 equilibrium θ values are relevant: $\theta_{\text{spl-ol}}$, $\theta_{\text{spl-opx}}$, $\theta_{\text{spl-cpx}}$, $\theta_{\text{ol-opx}}$, $\theta_{\text{ol-cpx}}$, and $\theta_{\text{opx-cpx}}$; all being a function of T . These individual θ values may vary for the same temperature. An example for a triangular relation instead of a linear relation is the 3-phase system CO_2 -water-vapor (Horita and Wesolowski, 1994; Meijer and

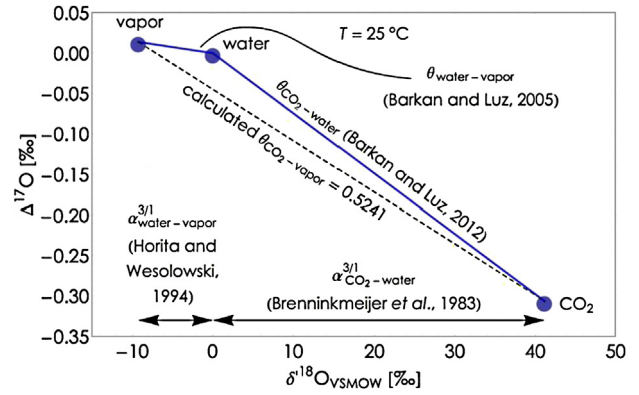


Fig. 5. Plot showing the equilibrium relations between CO_2 and water (Brenninkmeijer et al., 1983; Barkan and Luz, 2012) and water and vapor (Horita and Wesolowski, 1994; Barkan and Luz, 2005) at 25 °C (solid lines). The calculated $\theta_{\text{vapor-CO}_2}$ is shown. Note that the 3 coexisting phases are in isotope equilibrium, but do not fall on a common MFL.

Li, 1998; Barkan and Luz, 2005, 2012; Brenninkmeijer et al., 1983; Hofmann et al., 2012) (Fig. 5).

At the current level of precision, no variations in θ were identified for equilibrated rocks (Fig. 1A, C). In order to avoid conceptual pitfalls, one should, however, be aware that the rock mass fractionation line with >2 minerals is an approximation.

It is predicted from theory that equilibrium θ_{A-B} (Eq. (3)) decreases from 0.5305 with decreasing T (Matsuhisa et al., 1978; Young et al., 2002; Cao and Liu, 2011), but empirical equilibrium θ_{A-B} data are rare. To our knowledge only three published θ values exist for CO_2 -water (Hofmann et al., 2012; Barkan and Luz, 2012), vapor-water (Meijer and Li, 1998; Barkan and Luz, 2005) and apatite-water (Pack et al., 2013). In this study, we present empiric θ values for equilibrated high T rocks and low T minerals that precipitated from water (Figs. 1 and 2). Equilibration temperatures and water compositions are assumed as follows: The equilibration temperature for the San Carlos Iherzolite (1050 °C) is taken from Köhler and Brey (1990). The qz-plagioclase isotope thermometer (an77, Matsuhisa et al., 1979) gives 620 °C for the granite. The isotope temperature for the metamorphic BIF is 600 °C (Downs et al., 1981; Matthews et al., 1983; Chiba et al., 1989). For the chert, we assume that it precipitated from a mixture of 70% meteoric water with $\delta^{18}\text{O} = -6$ ‰ (falling on the MWL, Luz and Barkan, 2010) and 30% seawater (Barkan and Luz, 2011; Knauth, 1979; Fig. 2A) at a $T = 50 \pm 30$ °C (see thermometer by Kita et al., 1985). The mean growth temperature of the sponges spicules is estimated to 8 °C (see Table 1). The empiric relation between θ_{A-B} and T is displayed in Fig. 6.

The high- T rocks define MFLs with slopes of 0.528–0.529, which is close to the high- T approximation of 0.5305 (Matsuhisa et al., 1978; Young et al., 2002; Cao and Liu, 2011). At low temperature, silica-water, apatite-water (Pack et al., 2013), and CO_2 -water (Hofmann et al., 2012; Barkan and Luz, 2012) equilibrium slopes range between 0.521 and 0.523. For the silica-water equilibrium (constraint by the high- T approximation, chert-water, and sponge silica-water), our data suggest:

$$\theta_{\text{SiO}_2\text{-water}}^{\text{eq.}}(T) = \frac{-(740 \pm 290)}{T^2} + 0.5305 \quad (5)$$

This relation must be substantiated with more empiric data. The data for the silica-water equilibrium are clearly lower than predicted from theory by Cao and Liu (2011) (Fig. 6). The discrepancy may be due to approximations in the theoretical model or kinetic fractionation during chert and sponge silica precipitation. If the discrepancy were, indeed, due to kinetic effects, $\Delta^{17}\text{O}$ analysis would allow identification of phases that have not reached

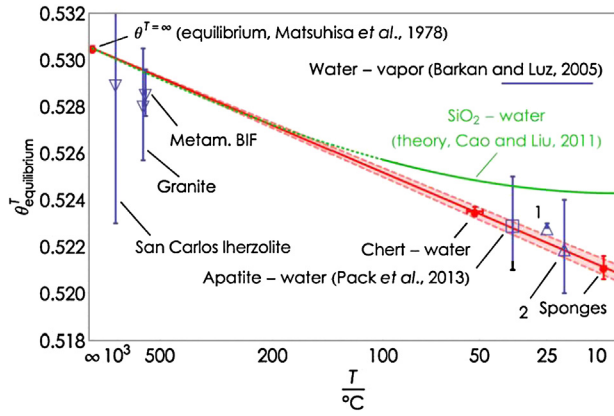


Fig. 6. Plot of θ vs. T^{-2} for equilibrium fractionation processes among minerals, minerals and fluids, and water and gases. The red line is the putative $\theta(T)$ relation for the silica–water equilibrium. The CO_2 –water (1: Barkan and Luz, 2012; 2: Hofmann et al., 2012, mean value) and apatite–water (Pack et al., 2013, corrected for the $\Delta^{17}\text{O}$ of NBS-28 from this study) $\theta(T)$ values coincide with the silica–water fractionation. High temperature rock assemblages have θ values that approach the high- T approximation of 0.5305. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

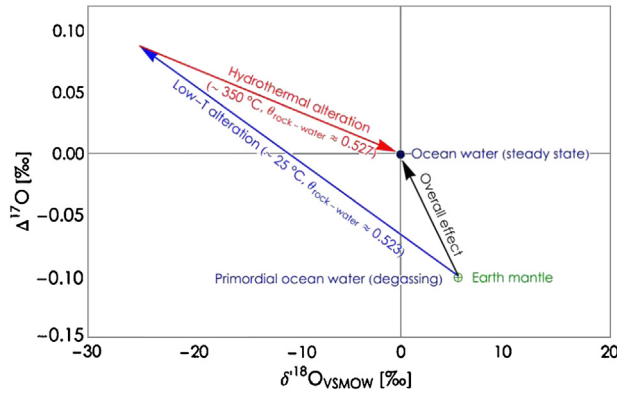


Fig. 7. Model explaining the oxygen triple isotope relation between the Earth mantle/oceanic crust and seawater. The difference in $\Delta^{17}\text{O}$ between mantle and modern ocean water is due to a combination of high- and low- T alteration of oceanic crust with different θ . Note that the arrows are low- and high- T isotope evolution directions of seawater and do not represent equilibrium lines.

equilibrium. In general, the uncertainty in θ is small for the low temperature equilibria because of the large $\Delta^{18}\text{O}$. This makes the T dependence of θ a potential new isotope thermometer for low- T systems.

An intriguing feature of the high precision $\Delta^{17}\text{O}$ data (Tanaka and Nakamura, 2013; this study) is the elevated $\Delta^{17}\text{O}$ of seawater relative to the Earth's mantle. The oxygen isotope composition of ocean water is buffered by low- and high- T isotopic exchange with mafic oceanic crust (Muehlenbachs and Clayton, 1976). Low- T ($\sim 25^\circ\text{C}$) exchange leads to formation of minerals with $\delta^{18}\text{O}$ that is $\sim 20\text{‰}$ higher than seawater (Savin and Epstein, 1970). Thus low- T alteration drives seawater to lower $\delta^{18}\text{O}$ (Muehlenbachs and Clayton, 1976). The low- T alteration is assigned a θ value of 0.523 (see Fig. 6). Hence, along with the decrease in $\delta^{18}\text{O}$, low- T alteration results in an increase in $\Delta^{17}\text{O}$ of ocean water. High- T exchange ($\sim 350^\circ\text{C}$) results in altered oceanic crust with $\delta^{18}\text{O} < 5.5\text{‰}$. A θ value of 0.527 is assumed for the high- T alteration (Fig. 6; Cao and Liu, 2011). High- T alteration leads to an increase in $\delta^{18}\text{O}$ and a decrease in $\Delta^{17}\text{O}$ of ocean water. Because the θ values for low- and high- T alteration are not identical, the effect for $\Delta^{17}\text{O}$ does not cancel out. Both, the $\delta^{18}\text{O}$ and $\Delta^{17}\text{O}$ of ocean water reflect the steady state between low- and high- T alteration (Fig. 7).

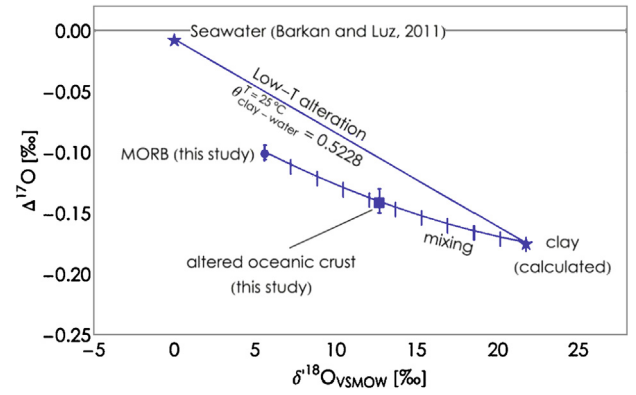


Fig. 8. Plot of $\Delta^{17}\text{O}$ vs. $\delta^{18}\text{O}$ illustrating the effect of low- T alteration on the composition of oceanic crust. The $\Delta^{17}\text{O}$ decreases due to formation of low- T alteration products with high $\delta^{18}\text{O}$ and $\theta = 0.523$ ($T = 25^\circ\text{C}$).

Our simplified model suggests that $\Delta^{17}\text{O}$ of ocean water is higher than the $\Delta^{17}\text{O}$ of the Earth mantle and fresh oceanic crust. This is, indeed, observed (Fig. 1). The slope of the line connecting oceanic crust and seawater of 0.512 (“Overall effect”, Fig. 7) has, as such, no physical meaning. It results from the combination of low- and high- T alteration. This agrees with classic models arguing that ocean water and oceanic crust are not in isotope equilibrium, but are in a steady state (Muehlenbachs and Clayton, 1976).

To test this simple model, we investigated the composition of fresh and altered oceanic crust. The $\Delta^{17}\text{O}$ of the mantle-derived ocean crust is -100 ± 6 ppm (MORB glass, this study). This is identical to the value of mantle xenoliths of -101 ± 2 ppm and is in agreement with no or only little fractionation between the Earth mantle and partial mantle melts (Eiler, 2001). A sample of altered oceanic crust has an elevated $\delta^{18}\text{O}$ by $\sim 7\text{‰}$ relative to unaltered crust (Table 2) that is accompanied by a marked decrease in $\Delta^{17}\text{O}$ by ~ 40 ppm (Fig. 8). The enrichment in ^{18}O is due to formation of low- T alteration products (e.g., clay; Savin and Epstein, 1970). The decrease in $\Delta^{17}\text{O}$ is explained by a combination of a high $\delta^{18}\text{O}_{\text{clay-water}}$ and a low clay–water equilibrium θ value. We assume that clay and seawater define a line with slope $\theta = 0.523$, which corresponds to an equilibration temperature of 25°C (Fig. 6). The composition of the altered oceanic crust sample falls on a mixing curve (clay, unaltered oceanic crust) and can well be explained by $\sim 45\%$ alteration with alteration products having a $\delta^{18}\text{O}$ value of 22‰ . The datum on the altered oceanic crust shows that low- T alteration leads, indeed, to a decrease in $\delta^{18}\text{O}$ and to an increase in $\Delta^{17}\text{O}$ of seawater.

Clearly, more data are required to affirm this model, but the data collected so far perfectly fit into the theoretical framework presented above. Both the elevated $\Delta^{17}\text{O}$ composition of seawater relative to the mantle and the $\Delta^{17}\text{O}$ composition of the altered oceanic crust agree with the simple steady state model. If confirmed this relation can be used to testify models suggesting that the ancient ocean water had lower $\delta^{18}\text{O}$ values (e.g., Wallmann, 2001) and that the ratio of low- to high- T alteration was different from today (e.g., Kamber, 2010).

5. Conclusions

- Gas chromatographic sample O_2 purification combined with long count times in the mass spectrometer results in an achievable precision of ± 5 ppm in $\Delta^{17}\text{O}$ for silicate and oxide samples. We report high-precision $\delta^{17}\text{O}_{\text{VSMOW}}$ values for reference materials (NBS-28 qz, UGW-2 gt, San Carlos ol).
- The $\Delta^{17}\text{O}$ of the Earth mantle has a $\Delta^{17}\text{O}$ of -101 ± 2 ppm relative to an RL with slope $\lambda_{\text{RL}} = 0.5305$ and zero intercept

(i.e., VSMOW $\in$ RL). The analysis of MORB glass suggests that fresh oceanic crust has a $\Delta^{17}\text{O}$ identical to the mantle value.

- Oxygen isotope data of meteorites and lunar samples must be corrected toward lower $\Delta^{17}\text{O}$ values if $\delta^{17}\text{O}$ was not reported on VSMOW scale. We present corrected literature data for HED, angrites, and Martian meteorites.
- Mineral assemblages in rocks fall on individual “rock” MFLs. These are characterized by individual slopes θ and intercepts. The concept of a single TFL should be abandoned.
- The equilibrium θ between minerals and minerals and fluids is a function of T with θ being close to 0.5305 at $>600^\circ\text{C}$ and about 0.5211 at $\sim 8^\circ\text{C}$. This relation has potential as low- T thermometer.
- The difference in $\Delta^{17}\text{O}$ between Earth mantle and seawater can be explained by a simple 2 step model with low- and high- T exchange between seawater and oceanic crust.

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